



BAULKHAM HILLS HIGH SCHOOL

2015 YEAR 11
YEARLY EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- In Questions 11–14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 9 pages.

This paper consists of TWO sections.

Section 1 – Page 2-4 (10 marks)

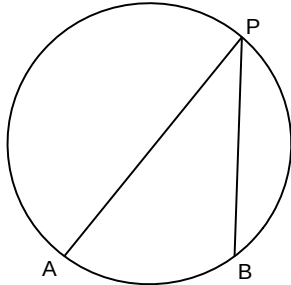
- Attempt Question 1-10
- Allow about **15** minutes for this section

Section II – Pages 5-9 (60 marks)

- Attempt questions 11-14
- Allow about **1 hours and 45** minutes for this section

Section 1 –Multiple Choice (10 marks)**Attempt all questions.****Answer the following in the booklet provided.**

- 1** AP is a diameter of the circle

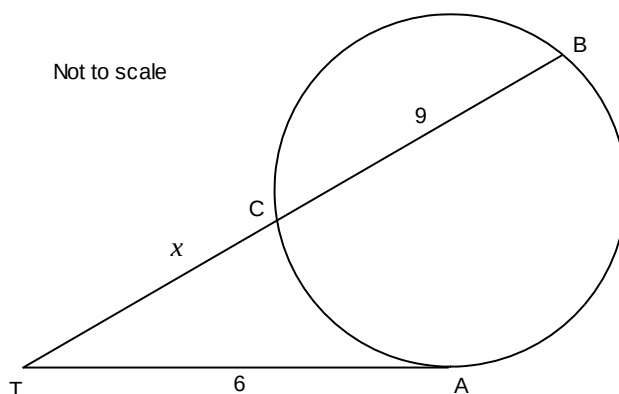


If $\angle APB = 40^\circ$, then $\angle PAB$ is

- (A) 40°
(B) 50°
(C) 60°
(D) not enough information
- 2** What is another expression for $\cos(x + y)$?
- (A) $\cos x \cos y - \sin x \sin y$
(B) $\cos x \cos y + \sin x \sin y$
(C) $\sin x \cos y - \cos x \sin y$
(D) $\sin x \cos y + \cos x \sin y$
- 3** What is the acute angle to the nearest degree between the lines $y = 1 - 3x$ and $4x - 6y - 5 = 0$?
- (A) 15°
(B) 38°
(C) 52°
(D) 75°
- 4** nC_r is equal to:
- (A) $(n - r)! {}^nP_r$
(B) $\frac{{}^nP_r}{(n-r)!}$
(C) $r! {}^nP_r$
(D) $\frac{{}^nP_r}{r!}$

5	Let $t = \tan \frac{\theta}{2}$ where $0^\circ < \theta < 180^\circ$ Which of the following gives the correct expression for $2 \sin \theta + 2 \cos \theta$? (A) $\frac{1+2t-t^2}{1+t^2}$ (B) $\frac{2+2t-2t^2}{1+t^2}$ (C) $\frac{2+4t-2t^2}{1+t^2}$ (D) $\frac{2-4t+2t^2}{1-t^2}$
6	The coordinates of a point that divides the interval $A(-2,7)$ to $B(12,0)$ externally in the ratio 4:3 is (A) $(-44, -28)$ (B) $(44, -28)$ (C) $(54, -21)$ (D) $(54, 21)$
7	If $\cos \theta = -\frac{3}{5}$ and $0^\circ < \theta < 180^\circ$ then $\tan \frac{\theta}{2}$ is equal to: (A) $-\frac{1}{3}$ or 3 (B) $\frac{1}{3}$ or 3 (C) -2 (D) 2
8	The expression $\sin x - \sqrt{3} \cos x$ can be written in the form $2 \sin(x + \beta)$. The value of β is (A) 30° (B) -30° (C) 60° (D) -60°

- 9** The line TA is a tangent to the circle at A , and TB is a secant meeting the circle at B and C .
Given that $TA = 6$, $CB = 9$ and $TC = x$, what is the value of x ?

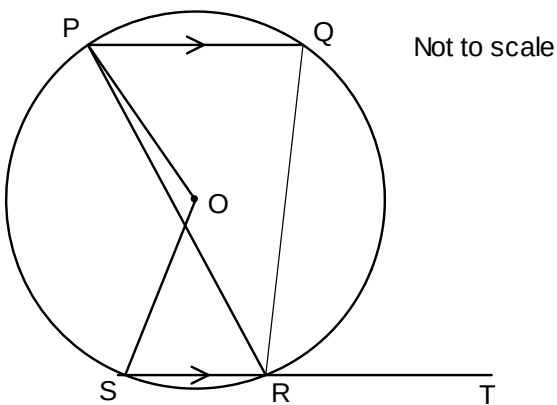


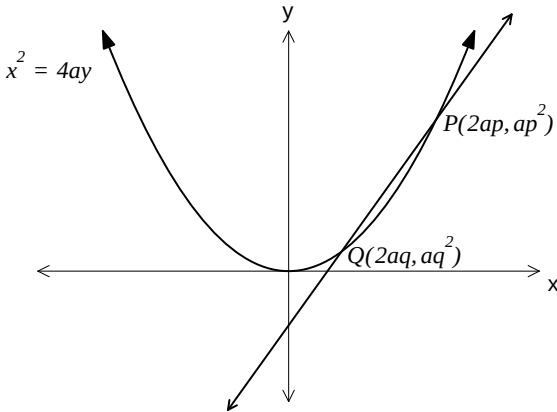
- (A) 2
(B) 3
(C) 12
(D) 18
- 10** A point moves in the xy -plane such that $P(\tan \theta, \cot \theta)$ is its parametric presentation with parameter θ , where θ is any real number. The locus of P is then a
- (A) Parabola
(B) Circle
(C) Hyperbola
(D) Straight line

End of Section I

Section II – Extended Response

All necessary working should be shown in every question.

Question 11 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Solve $\frac{2x - 3}{x - 2} \geq 1$	3
b)	The acute angle between the lines $y = (m + 3)x$ and $y = mx$ is 45° (i) Show that $\left \frac{3}{m^2 + 3m + 1} \right = 1$ (ii) Hence find all the values of m.	1 2
c)	In the diagram, PQ is parallel to ST and O is the centre of the circle. $\angle PQR = 80^\circ$ and $\angle POS = 110^\circ$  Not to scale Copy the diagram into your booklet. Find the value of $\angle PRQ$ giving reasons.	3
d)	Solve $5 \cos \theta + 12 \sin \theta - 13 = 0$ over the domain $0^\circ \leq \theta \leq 360^\circ$.	3
e)	How many distinct permutations of the letters of the word $DIVIDE$ are possible in a straight line when the word begins and ends with the letter D ?	1
f)	If $\tan(A + B) = x$ and $\tan B = \frac{1}{3}$, express $\tan A$ in terms of x .	2
End of Question 11		

Question 13 (15 marks) - Start on the appropriate page in your answer booklet	Marks
a) 4 men, 2 women and a child sit at a round table. In how many ways can these 7 people be arranged (i) Without restrictions. (ii) If the child is seated between the two men.	1 1
b) Consider the function $f(x) = \frac{x}{4-x^2}$ (i) State the domain of the function. (ii) Show that the function is an odd function. (iii) Show that the function is increasing throughout its domain. (iv) Evaluate $\lim_{x \rightarrow \infty} f(x)$. (v) Using the above information sketch the function $y = f(x)$ showing any essential features.	1 2 2 1 2
c) The diagram below shows the parabola $x^2 = 4ay$ and the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$  (i) Show that the equation of the chord PQ is $2y = (p + q)x - 2apq$. (ii) The line joining P and Q passes through the point $(0, -2a)$. Show that $pq = 2$. (iii) The normals to the parabola $x^2 = 4ay$ at points P and Q intersect at K. The coordinates of K are $(-apq(p + q), a(p^2 + q^2 + pq + 2))$ (DO NOT PROVE THIS) Find the equation of the locus of K.	2 1 2
End of Question 13	

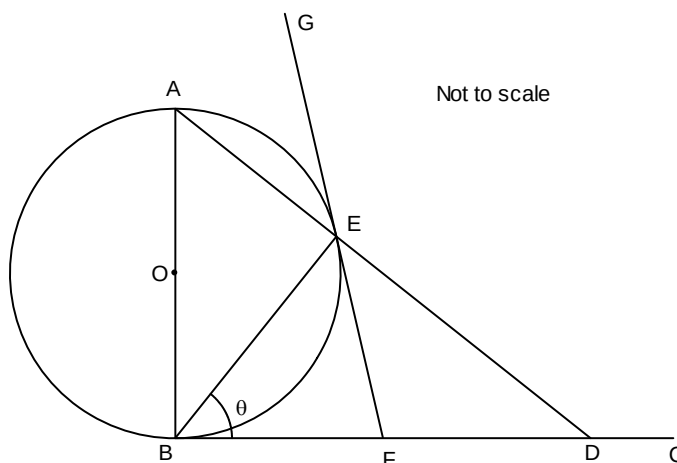
Question 14 (15 marks) - Start on the appropriate page in your answer booklet

a) Simplify fully:

$$\cos^2 A + \cos^2(120^\circ + A) + \cos^2(120^\circ - A).$$

3

b) In the diagram, AB is the diameter of the circle centre O , and BC is a tangent at B . The line AD intersects the circle at E and BC at D . The tangent to the circle at E intersects BC at F . Let $\angle EBF = \theta$.



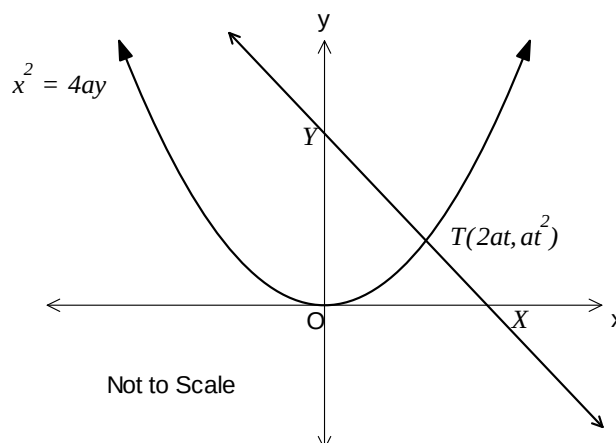
(i) Prove $\angle FED = 90^\circ - \theta$.

2

(ii) Prove $BF = FD$.

2

c) In the diagram, $T(2at, at^2)$ is a point on the parabola $x^2 = 4ay$



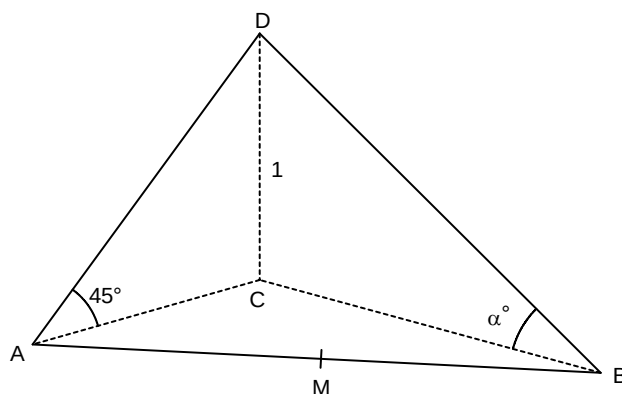
(i) Show that the normal to the parabola at T has equation $x + ty = 2at + at^3$.

2

(ii) This normal cuts the x and y axes at X and Y respectively.
Find the ratio of $\frac{TX}{TY}$ in simplest terms.

2

- d) CD is a vertical pole of height 1 metre that stands with its base C on horizontal ground. A is a point due South of C such that the angle of elevation of D from A is 45° . B is a point due East of C such that the angle of elevation of D from B is α . M is the midpoint of AB .



(i) Show that $AB = \operatorname{cosec} \alpha$.

2

(ii) Find an expression for CM in terms of $\operatorname{cosec} \alpha$.

2

End of Examination

13 a	For which x - values on the curve $y = \frac{x}{x+1}$ is the curve increasing?	2
13 b	(i) If $a = \frac{15bx}{3b+5x}$ express x in terms of a and b	2
	Hence express $\sqrt{\frac{3b-a}{5x-a}}$ in terms of a and b	2
	(ii)	

Answers

multiple choice

1. B (L in a semicircle)

2. A

$$3. D. \quad \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-3 - \frac{2}{3}}{1 + -3 \times \frac{2}{3}} \right|$$

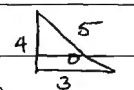
$$\theta = 75^\circ \text{ (nearest degree)}$$

4. D.

$$5. C \quad \frac{2 \times 2t}{1+t^2} + \frac{2 \times (1-t^2)}{1+t^2} = \frac{4t+2-2t^2}{1+t^2}$$

$$6. C \quad \left(\frac{mx_2 - ny_1}{m-n}, \frac{my_2 - nx_1}{m-n} \right) \quad m:n = 4:3$$

$$\left(\frac{4 \times 2 - 3 \times 2}{4-3}, \frac{4 \times 0 - 3 \times 7}{4-3} \right) = (54, -21)$$

~~7. D.~~  2nd quad. $\tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$

(Answer missing) $-\frac{4}{3} = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$

$$(2 \tan \frac{\theta}{2} + 1)(\tan \frac{\theta}{2} - 2) = 0$$

$$\tan \frac{\theta}{2} = 2$$

$$8. D. \quad \tan \beta = \frac{\sqrt{3}}{1}$$

$$\beta = 60^\circ \text{ but } (\alpha + \beta) \text{ should be } (\pi - \beta)$$

$$\therefore \beta = -60^\circ$$

$$9. B. \quad AT^2 = TB \times TC$$

$$b^2 = (x+9)x$$

$$10. C. \quad x = \tan \theta \quad y = \cot \theta$$

$$= \frac{1}{\tan \theta}$$

$$= \frac{1}{x}$$

$$\therefore xy = 1$$

10 marks

Question 11.

$$a) \frac{2x-3}{x-2} \geq 1$$

$$\frac{2x-3-x+2}{x-2} \geq 0$$

① identifies $x=2$, 1

① correct answer

$$(x-2) \times \frac{(x-1)}{x-2} \geq 0 \times (x-2)$$

① $x \neq 2$

$$(x-2)(x-1) \geq 0$$

$$\therefore x \leq 1 \text{ or } x \geq 2 \text{ but } x \neq 2 \therefore x \leq 1 \text{ or } x \geq 2$$

$$b) l_1: m_1 = m+3$$

$$l_2: m_2 = m$$

$$\theta = 45^\circ$$

$$i) \quad \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45 = \left| \frac{m+3-m}{1+(m+3)m} \right|$$

① should have

either -1st 2 lines

$$1 = \left| \frac{3}{m^2+3m+1} \right|$$

ii) two cases

① for solving absolute value equations

$$m^2+3m+1=3$$

$$m^2+3m+1=-3$$

$$m^2+3m-2=0$$

$$m^2+3m+4=0$$

$$m = \frac{-3 \pm \sqrt{9+8}}{2}$$

$$m = \frac{-3 \pm \sqrt{9-16}}{2}$$

$$= \frac{-3 \pm \sqrt{17}}{2} \quad \text{①}$$

$\therefore$ no soln

the soln

3 marks

$$c) \angle PRS = \frac{1}{2} \angle POS \text{ (L at centre twice L at circumference on same arc)}$$

$$= 55^\circ \quad \text{①}$$

$$\angle QRT = \angle PQR \text{ (alternate Ls, } PQ \parallel ST)$$

$$= 80^\circ \quad \text{①}$$

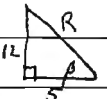
$$\angle PRQ + \angle PRS + \angle QRT = 180^\circ \text{ (L sum straight line)}$$

$$\therefore \angle PRQ = 180 - (55 + 80) \quad \text{①}$$

$$= 45^\circ$$

d) $5 \cos \theta + 12 \sin \theta = 13$ $0 \leq \theta \leq 360^\circ$ (3 marks)

$5 \cos \theta + 12 \sin \theta = R \cos(\theta - \beta)$

where R :  $R = \sqrt{144 + 25} = 13$ $\tan \beta = \frac{12}{5}$ $\beta = 67^\circ 23'$ (1)

state if corrected off?

$\therefore 13 \cos(\theta - 67^\circ 23') = 13$ (1) $0 \leq \theta - 67^\circ 23' \leq 292^\circ 37'$

$\theta - 67^\circ 23' = 0, 360^\circ$ not in range

$\theta = 67^\circ 23'$ (1)

OK.
t method:

$\frac{5(1-t^2)}{1+t^2} + \frac{12 \times 2t}{1+t^2} = 13$ where $t = \tan \frac{\theta}{2}$

$5 - 5t^2 + 24t = 13 + 13t^2$ (1)

$18t^2 - 24t + 8 = 0$

$9t^2 - 12t + 4 = 0$

$(3t - 2)^2 = 0$

$t = \frac{2}{3}$ $\tan \frac{\theta}{2} = \frac{2}{3}$

$\frac{\theta}{2} = 33.69^\circ$ (1)

$\theta = 67^\circ 23'$ (1)

test 180°

LHS = $5 \cos 180 + 12 \sin 180$

$= -5$

$\neq 13$ $\therefore$ no soln $\therefore \theta = 67^\circ 23'$

e. DIVIDE

$\boxed{D} \text{ --- } \boxed{D} \therefore \text{no. of ways} = \frac{4!}{2!}$

$= 12$ (1)

f. $\tan(A+B) = x$ $\tan B = \frac{1}{3}$

$\therefore \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$x = \frac{\tan A + \frac{1}{3}}{1 - \frac{1}{3} \tan A}$ (1)

$\frac{3x - x \tan A}{3} = \frac{3 \tan A + 1}{3}$

$3x - 1 = 3 \tan A + x \tan A$

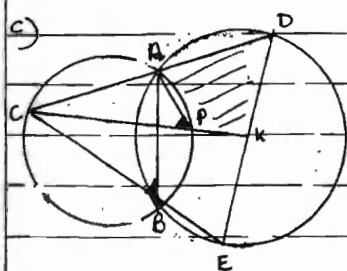
$\tan A = \frac{3x - 1}{3 + x}$ (1)

/15

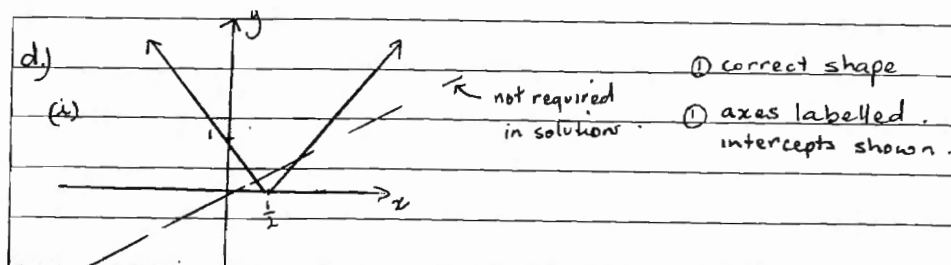
Question 12.

a) $\frac{4^n + 8(2^n)}{2^{n+1}} = \frac{2^{2n} + 2^3 \cdot 2^n}{2^n \cdot 2^{n+1}}$ ①
 $= \frac{2^n(2^n + 2^3)}{2^n \cdot 2^{n+1}}$
 $= 2^{n+1} + 2^4 \text{ or } 2^{n+1} + 16.$ ①

b) i) no. of ways = ${}^7C_3 \times {}^6C_2$ ① recognising
 $= 525$ nC_r
 ii) boys together = $2!$ ① answer
 no. of ways in a line = $4! \times 2!$ ① putting boys
 $= 48$ together
 ① answer



i) $\angle APC = \angle ABC$ ①
 (Ls at circumference equal on same arc/chord)
 ii) In cyclic quad ADEB
 $\angle ADE = \angle ABC$ (external L equal interior opposite L) ①
 but $\angle ABC = \angle APC$ (part i)
 and $\angle APC = \angle ADK = \angle ABC$
 $\therefore$ external L equals interior opposite L ①
 $\therefore$ ADKP is a cyclic quadrilateral



ii) $1 - 2x \leq x$ $-1 + 2x \leq x$
 $1 \leq 3x$ $-1 \leq -x$
 $x \geq \frac{1}{3}$ $x \leq 1$ ① correct values
 $\therefore \frac{1}{3} \leq x \leq 1$ ① put together in one inequality

e) Prove: $\cos 2\theta + \tan \theta \sin 2\theta = 1$
 LHS = $2\cos^2 \theta - 1 + \frac{\sin \theta}{\cos \theta} \cdot 2\sin \theta \cos \theta$ ①
 $= 2\cos^2 \theta + 2\sin^2 \theta - 1$
 $= 2(\cos^2 \theta + \sin^2 \theta) - 1$ ①
 $= 1$
 $= \text{RHS.}$

Question 13

BOS#:

a) 4 men 2 women 1 child = 7

i) no. of ways = 6!

$$= 720$$

①

ii) no. of ways = ${}^4C_2 \times 2! \times 4!$

$$= 288$$

①

b) $f(x) = \frac{x}{4-x^2}$

→ D: all real x , $x \neq \pm 2$

①

ii) $f(x) = \frac{x}{4-x^2}$

$$f(-x) = \frac{-x}{4-(-x)^2}$$

① should show substitute

$$= \frac{-x}{4-x^2} = -f(x) \therefore \text{odd function. } \textcircled{1} \text{ concluding}$$

iii) $f(x) = \frac{x}{4-x^2}$

$$f'(x) = \frac{(4-x^2) \cdot 1 - x(-2x)}{(4-x^2)^2}$$

① $f'(x)$

$$= \frac{4+x^2}{(4-x^2)^2}$$

as numerator and denominator are > 0 for all values of x

① concluding

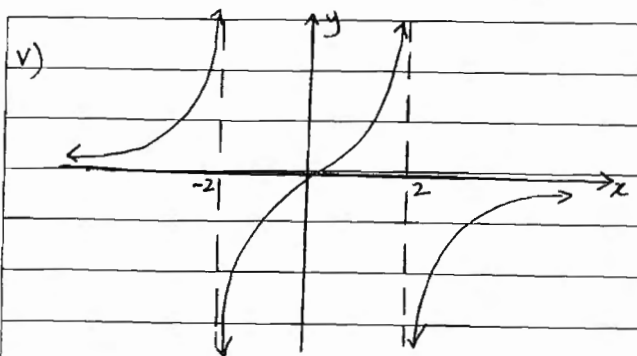
$\therefore f(x) = y$ is always increasing for its domains.

$$\lim_{x \rightarrow \infty} \frac{x}{4-x^2} = \lim_{x \rightarrow \infty} \frac{\frac{x}{x^2}}{\frac{4}{x^2} - \frac{x^2}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{4}{x^2} - 1}$$

①

$= 0 \therefore$ curve approaches 0 as $x \rightarrow \infty$



① asymptotes and centre

①: curve/graph.

c) i) PA: $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{ap^2 - aq^2}{2ap - 2aq}$$

①

$$= \frac{p+q}{2}$$

$$y - y_1 = m(x - x_1) \text{ using } P(2ap, ap^2)$$

$$y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$$

$$y - ap^2 = \left(\frac{p+q}{2}\right)x - \frac{2ap^2}{2} - \frac{2apq}{2}$$

①

$$y = \left(\frac{p+q}{2}\right)x - apq$$

$$\therefore 2y = (p+q)x - 2apq$$

ii) sub in $(0, -2a)$

$$\therefore -4a = 0 - 2apq$$

①

$$pq = 2$$

$$\text{iii) } x = -apq(p+q)$$

$$= -2a(p+q)$$

$$p+q = \frac{x}{-2a}$$

①

$$y = a(p^2 + q^2 + pq + 2)$$

$$= a((p+q)^2 - pq + 2)$$

$$= a\left(\left(\frac{-x}{2a}\right)^2 - 2 + 2\right)$$

$$= \frac{x^2}{4a}$$

$$\therefore x^2 = 4ay$$

①

Question 14

BOS#:

$$\begin{aligned}
 a) \quad & \cos^2 A + \cos^2 (120^\circ + A) + \cos^2 (120^\circ - A) \\
 &= \cos^2 A + \frac{1}{2} (\cos 2(120^\circ + A) + 1) + \frac{1}{2} \cos (2(120^\circ - A) + 1) \\
 &= \cos^2 A + \frac{1}{2} \cos (240^\circ + 2A) + \frac{1}{2} + \frac{1}{2} \cos (240^\circ - 2A) + \frac{1}{2} \\
 &= 1 + \cos^2 A + \frac{1}{2} (\cos 240^\circ \cos 2A - \sin 240^\circ \sin 2A) \\
 &\quad + \frac{1}{2} (\cos 240^\circ \cos 2A + \sin 240^\circ \sin 2A) \\
 &= 1 + \cos^2 A + \cos 240^\circ \cos 2A \quad (\cos 240^\circ = -\frac{1}{2}) \\
 &= 1 + \cos^2 A - \frac{1}{2} (2 \cos^2 A - 1) \\
 &= 1 + \cos^2 A - \cos^2 A + \frac{1}{2} \\
 &= \frac{3}{2}
 \end{aligned}$$

② working towards... many methods

$$\left[\frac{3}{2} \cos^2 x \right] \text{ ① mark only}$$

① final answer

b. i) Aim: prove $\angle FED = 90^\circ - \theta$ ②

$\angle AEB = 90^\circ$ (L in a semicircle)

$EF = FB$ (tangents from an external point are equal)

$\therefore \angle FBE = \angle FEB$ (Ls opposite equal sides are equal)

$\therefore \hat{AEB} + \hat{BEF} + \hat{FED} = 180^\circ$ (L sum straight line)

$\angle FED = 180 - 90 - \theta$

$= 90 - \theta$

ii) Prove: $BF = FD$.

$\angle EFD = 2\theta$ (external L of $\triangle BEF$ equal to interior opposite Ls) ②

then $\angle FDE = 180 - \hat{EFD} - \hat{FED}$ (L sum $\triangle$)

$= 180 - (90 - \theta) - 2\theta$

$= 90 - \theta$

$$\therefore \hat{FDE} = \hat{FED}$$

and $FE = FD$ (sides opposite equal Ls are equal)

but $BF = FE$ (proved above)

$$\therefore \underline{BF = FD}$$

c. i) $\ddot{x} = 4ay$ at T ($2at, at^2$)

$$y = \frac{\ddot{x}}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

① establishing gradient

at $x = 2at$ $m_1 = x$ $m_2 = -\frac{1}{m_1}$

$$= -\frac{1}{x}$$

normal: $y - y_1 = m(x - x_1)$

$$y - at^2 = -\frac{1}{x}(x - 2at)$$

① this line

$$xy - at^3 = -x + 2at$$

$$x + xy = 2at + at^3$$

ii) at X $y = 0 \therefore x = 2at + at^3$ ($2at + at^3, 0$)

at Y $x = 0 \therefore y = 2a + at^2$ ($0, 2a + at^2$)

length TY = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$= \sqrt{(2at)^2 + (at^2 - 2a - at^2)^2}$$

$$= \sqrt{4a^2t^2 + 4a^2}$$

$$= 2a\sqrt{t^2 + 1}$$

(similarly)

length TX = $\sqrt{(2a + at^2 - 2at)^2 + (0 - at^2)^2}$

$$= \sqrt{a^2t^4 + a^2t^4}$$

$$= at^2\sqrt{t^2 + 1}$$

① method. ratio distance?

$$\therefore \frac{TX}{TY} = \frac{at^2\sqrt{t^2 + 1}}{2a\sqrt{t^2 + 1}}$$

$$= \frac{t^2}{2}$$

① $\frac{t^2}{2}$

d) in $\triangle ADC$

$$\tan 45 = \frac{DC}{AC}$$

$$AC = 1$$

in $\triangle BDC$

$$\tan \alpha = \frac{DC}{CB}$$

$$CB = \frac{1}{\tan \alpha} = \cot \alpha$$

① establish $\triangle$'s.

$$\textcircled{1} AB^2 = 1 + \cot^2 \alpha$$

i) now $\hat{ACB} = 90^\circ$

in $\triangle ACB$

$$AC^2 + CB^2 = AB^2 \quad (\text{pythagoras' theorem})$$

$$\therefore AB^2 = 1 + \cot^2 \alpha$$

$$= \operatorname{cosec}^2 \alpha$$

$$\therefore AB = \operatorname{cosec} \alpha$$

ii) since $\hat{ACB} = 90^\circ$

and m is the midpoint of AB.

then $\hat{ACB}$ is the $\angle$ in a semicircle through

A, C, B with diameter AB.

① for valid method.

$$\therefore CM = \frac{1}{2} AB$$

$$= \frac{1}{2} \operatorname{cosec} \alpha$$

$$\textcircled{1} \frac{1}{2} \operatorname{cosec} \alpha$$

[①/2 for assuming $\alpha = 45$ and achieving correct answer]

/15